2.1 Functions of an Academic Advising Chatbot

2.1.1 Responding to Student Queries

One of the primary functions of an academic advising chatbot is to respond to student queries in real time. Chatbots in higher education are increasingly used to provide instant answers to frequently asked questions related to registration, course selection, academic policies, and administrative procedures (Okonkwo & Ade-Ibijola, 2021; Winkler & Söllner, 2018). By leveraging Natural Language Processing (NLP), these systems can interpret user intent and generate contextually relevant responses, thereby reducing the workload of human advisors while maintaining accessibility (Følstad & Brandtzaeg, 2017; Kerly, Hall, & Bull, 2007). Research further shows that conversational agents improve response efficiency and enhance student engagement when compared to static FAQ systems (Bii, Too, & Langat, 2013; Clarizia et al., 2018).

Several implemented chatbot systems in higher education demonstrate the effectiveness of automated query response. For instance, Kerly, Hall, and Bull (2007) developed a conversational agent to support student learning and advising, showing that dialogue-based systems can answer structured academic questions while reducing advisor workload. Similarly, Clarizia et al. (2018) designed an educational support chatbot that responded to student inquiries regarding courses, schedules, and institutional procedures, reporting improved accessibility and faster response times.

More recently, Okonkwo and Ade-Ibijola (2021), in their systematic review of educational chatbots, identified query handling as the most common implemented function in academic support systems. Additionally, Winkler and Söllner (2018) evaluated chatbot adoption in universities and found that conversational systems successfully automated frequently asked academic questions while maintaining high perceived usefulness among students.

These implementations confirm that automated query response is a core and validated function of academic advising chatbots. Therefore, incorporating real-time query handling into the proposed system aligns with established system designs in higher education. Therefore, for the proposed Academic Advising Chatbot, this function is critical because it ensures that students receive immediate guidance without waiting for physical appointments, particularly during peak academic periods.

2.1.2 Providing Academic Progress Tracking

Another important function of academic advising systems is the ability to track and monitor students' academic progress. Intelligent advising systems can integrate with institutional databases to analyze completed units, grades, credit accumulation, and graduation requirements (Pardo et al., 2019; Ifenthaler & Yau, 2020). Learning analytics techniques enable such systems to identify at-risk students and provide personalized recommendations for improvement (Ferguson, 2012; Siemens & Baker, 2012).

Studies indicate that data-driven academic support systems significantly improve student retention and academic performance when progress tracking is automated and continuously monitored (Arnold & Pistilli, 2012; Campbell, DeBlois, & Oblinger, 2007).

Academic progress tracking has been implemented in several intelligent advising and learning analytics systems. For instance, Arnold and Pistilli (2012) developed the Course Signals system at Purdue University, which monitored student performance data and provided early feedback to support academic success. Their findings demonstrated that automated progress monitoring significantly improved student retention rates. Similarly, Campbell, DeBlois, and Oblinger (2007) described academic analytics systems that integrate institutional data to evaluate student performance and guide academic decision-making.

In addition, Pardo et al. (2019**)** implemented a personalized feedback system using learning analytics dashboards to track student engagement and performance in real time. Likewise, Ifenthaler and Yau (2020) demonstrated how analytics-based advising systems provide predictive insights that support early intervention strategies.

These implemented systems confirm that integrating academic performance tracking into advising technologies enhances personalized support. Therefore, embedding academic progress tracking within the proposed Academic Advising Chatbot will allow it to provide tailored academic guidance rather than generic information.

2.1.3 Sending Deadline Reminders

Automated reminder systems are another essential function of academic advising chatbots. Research shows that timely nudges and reminders significantly influence student behavior, particularly in relation to registration deadlines, fee payments, and assignment submissions (Castleman & Page, 2015; Fryer, 2017). Behavioral intervention studies demonstrate that SMS-based and chatbot-based reminders reduce missed deadlines and improve enrollment persistence (Bird et al., 2019; Page & Gehlbach, 2017).

Automated reminder functionality has been successfully implemented in student support systems to improve academic compliance and persistence. For example, Castleman and Page (2015) implemented an SMS-based reminder system that provided students with timely nudges about enrollment and financial aid deadlines, resulting in improved college enrollment rates. Similarly, Bird et al. (2019) evaluated large-scale reminder campaigns and found that automated messaging significantly reduced missed administrative deadlines.

In institutional settings, Page and Gehlbach (2017) implemented behavioral “nudge” interventions delivered through digital communication platforms, demonstrating measurable improvements in student task completion. Additionally, Kuh et al. (2008) emphasized that proactive communication systems enhance student engagement and institutional effectiveness.

These implementations demonstrate that automated reminders are not merely theoretical features but empirically validated functions in academic support systems. In higher education settings, reminder systems integrated into digital advising platforms enhance communication efficiency and

reduce administrative bottlenecks (Kuh et al., 2008; Tinto, 2012). Incorporating automated deadline notifications into the proposed Academic Advising Chatbot will therefore ensure proactive communication, helping students stay informed and academically compliant.

2.2 Features of the Smart Campus Navigation System

2.2.1 Real-Time Location and Navigation Updates

Real-time location tracking is a core feature of smart campus navigation systems. This feature enables the system to determine a user's current position and dynamically update routes as they move across the campus. Laoudias et al. (2018) implemented a real-time indoor positioning system that combined Wi-Fi fingerprinting and signal processing to improve location accuracy. Similarly, Zafari, Gkelias, and Leung (2019) reviewed implemented indoor positioning systems that use Bluetooth Low Energy (BLE) beacons and Wi-Fi triangulation to deliver continuous location updates. These systems demonstrate that real-time navigation relies on sensor-based data acquisition and algorithmic route recalculation.

In university contexts, Cheng et al. (2016) developed a campus navigation system integrating GPS for outdoor navigation and Wi-Fi-based indoor positioning for building-level guidance. Their system recalculated routes automatically when users deviated from suggested paths. Likewise, Li, Chen, and Pei (2017) implemented a hybrid campus navigation application that seamlessly switched between GPS and indoor positioning systems, thereby ensuring uninterrupted guidance. These implementations confirm that real-time updates are technically feasible and practical in academic environments.

The importance of real-time updates lies in their ability to enhance user experience and reduce navigation errors. Static maps do not adjust to movement or changes in environment, whereas dynamic navigation systems improve accuracy and efficiency (Zafari et al., 2019; Cheng et al., 2016). Real-time functionality is particularly important for new students who may be unfamiliar with campus layout and require step-by-step directions. Additionally, this feature supports time-sensitive movement between lecture halls, offices, and laboratories.

However, real-time navigation systems face notable challenges. Indoor positioning technologies may suffer from signal interference, infrastructure costs, and accuracy limitations (Laoudias et al., 2018; Zafari et al., 2019). Wi-Fi fingerprinting requires calibration and maintenance, while Bluetooth beacon deployment increases operational expenses. These limitations suggest that while real-time updates significantly enhance usability, the proposed Smart Campus Navigation System must balance accuracy, infrastructure cost, and scalability when implementing this feature.

2.2.2 Search and Location-Based Query Functionality

Search functionality enables users to locate buildings, lecture halls, departments, or facilities by entering keywords into a search interface. In implemented navigation systems, this feature is supported by indexed campus databases and geospatial querying mechanisms. For example, Kim, Kim, and Yoo (2014) developed a mobile campus map system that allowed users to search locations by name or category, retrieving coordinates from a centralized spatial database. Similarly, Borrmann et al. (2015) implemented an indoor navigation application where users could query rooms and offices using structured search filters linked to building information models.

In addition, Huang, Gartner, and Krisp (2018) demonstrated that geocoded search systems enhance wayfinding performance by reducing cognitive load during navigation tasks. Their system integrated semantic search capabilities to allow users to search for functional categories such as "library," "finance office," or "computer lab." Likewise, Li et al. (2017) incorporated keyword-based querying in their hybrid campus navigation system, enabling quick identification of facilities within large academic compounds.

The importance of search functionality lies in its support for usability and efficiency. Without search capability, users must manually browse maps, which increases navigation time and cognitive effort (Huang et al., 2018; Kim et al., 2014). A well-designed search feature improves accessibility, especially in large campuses with complex building structures. It also supports inclusive design by assisting users who may not be familiar with spatial layouts or building names.

Despite its advantages, search functionality may present limitations if poorly implemented. Inaccurate database indexing or incomplete spatial data may produce incorrect or empty results (Borrmann et al., 2015; Huang et al., 2018). Furthermore, simple keyword-based search systems may fail to interpret ambiguous queries, thereby reducing user satisfaction. Advanced semantic search mechanisms improve accuracy but require more computational resources and structured metadata. Therefore, the proposed Smart Campus Navigation System should incorporate a well-maintained geospatial database and user-friendly search interface to ensure reliability and efficiency.

2.3 Components of the Smart Campus Navigation System

2.3.1 Location Detection Module

The Location Detection Module is a core component of a smart campus navigation system, responsible for determining the real-time position of users within the campus environment. In outdoor environments, this module typically relies on Global Positioning System (GPS) technology, while indoor positioning requires alternative techniques such as Wi-Fi fingerprinting, Bluetooth Low Energy (BLE) beacons, or Radio Frequency Identification (RFID). Cheng et al. (2016) implemented a hybrid campus navigation system combining GPS and Wi-Fi-based indoor positioning to enable seamless transition between outdoor and indoor navigation. Similarly, Zafari, Gkelias, and Leung (2019) reviewed practical implementations of indoor localization systems that utilize BLE beacons and signal strength triangulation to improve positioning accuracy.

The role of this component is to continuously capture spatial coordinates and transmit them to the routing engine for path computation. Without accurate location detection, route guidance becomes unreliable. Laoudias et al. (2018) demonstrated that Wi-Fi fingerprinting techniques significantly enhance indoor positioning accuracy in complex environments such as university campuses. Likewise, Li, Chen, and Pei (2017) developed a campus navigation application that dynamically adjusted user location based on sensor fusion techniques combining GPS and inertial sensors. These implementations confirm that reliable location detection is foundational for effective navigation.

The importance of this module lies in its direct impact on usability and system credibility. If the system inaccurately identifies user location, it may generate incorrect routes, thereby reducing user trust. Research shows that positioning accuracy below a certain threshold significantly affects user satisfaction in navigation systems (Zafari et al., 2019; Laoudias et al., 2018). Therefore, robust localization methods are essential in educational institutions where buildings may have similar layouts or closely spaced lecture rooms.

For the proposed Smart Campus Navigation System at ABC University, the Location Detection Module will integrate GPS for outdoor movement and Wi-Fi-based positioning for indoor environments. Given possible infrastructure limitations, a hybrid approach will balance accuracy and cost-effectiveness. By aligning with approaches used by Cheng et al. (2016) and Li et al. (2017), the system can ensure seamless navigation across lecture halls, administrative offices, and laboratories, thereby enhancing student mobility across campus.

2.3.2 Routing and Pathfinding Engine

The Routing and Pathfinding Engine is the component responsible for computing optimal routes between a user's current location and their desired destination. This module typically implements graph-based algorithms such as Dijkstra's algorithm or A* (A-star) search to determine the shortest or fastest path. Delling et al. (2009) implemented advanced route planning algorithms in large-scale navigation systems to improve computational efficiency. In academic environments, Cheng et al. (2016) applied graph-based shortest path algorithms in their campus navigation prototype to provide real-time route recommendations.

The primary role of the routing engine is to process spatial data received from the location detection module and compute navigable paths based on predefined campus maps. Borrmann et al. (2015) developed an indoor navigation system integrating building information modeling (BIM) data with routing algorithms to compute accessible paths within complex structures. Similarly, Li et al. (2017) implemented dynamic route recalculation when users deviated from suggested paths, demonstrating the importance of adaptive routing in real-world scenarios.

The significance of the routing engine lies in its influence on navigation efficiency and responsiveness. Efficient algorithms reduce computation time, especially when multiple users access the system simultaneously. Research indicates that heuristic-based algorithms such as A* improve speed while maintaining path optimality (Delling et al., 2009; Borrmann et al., 2015).

However, routing systems may face limitations if campus maps are not accurately modeled or regularly updated, leading to outdated or inaccessible paths.

Within the proposed Smart Campus Navigation System for ABC University, the Routing and Pathfinding Engine will utilize a graph representation of campus buildings and walkways. Nodes will represent buildings and rooms, while edges will represent walkable paths. By implementing a computationally efficient shortest-path algorithm, the system will provide accurate and responsive navigation guidance. Drawing from implementations by Cheng et al. (2016) and Li et al. (2017), the routing engine will also support dynamic route recalculation, ensuring adaptability when users change direction or encounter obstacles.

2.4 Types of Smart Campus Navigation Systems

Smart campus navigation systems can be classified into GPS-based outdoor systems and Indoor Positioning Systems (IPS)**.** Each type has unique technological requirements, advantages, and limitations, which influence their suitability for campus environments. The classification helps developers select the appropriate approach for system design and implementation.

2.4.1 GPS-Based Navigation Systems

GPS-based navigation systems rely on satellite signals to determine a user's geographical location. They are widely used in outdoor navigation applications and have been successfully applied in university campuses for routing between buildings. Cheng et al. (2016) developed a hybrid campus navigation system that utilized GPS for outdoor navigation, demonstrating high positional accuracy for guiding students across large open spaces. Similarly, Kim, Kim, and Yoo (2014) implemented a GPS-based campus mapping application for Seoul National University, which enabled students to locate lecture halls, cafeterias, and administrative offices efficiently. Figure 2.1 below shows google maps which is an example of GPS- Based navigation system

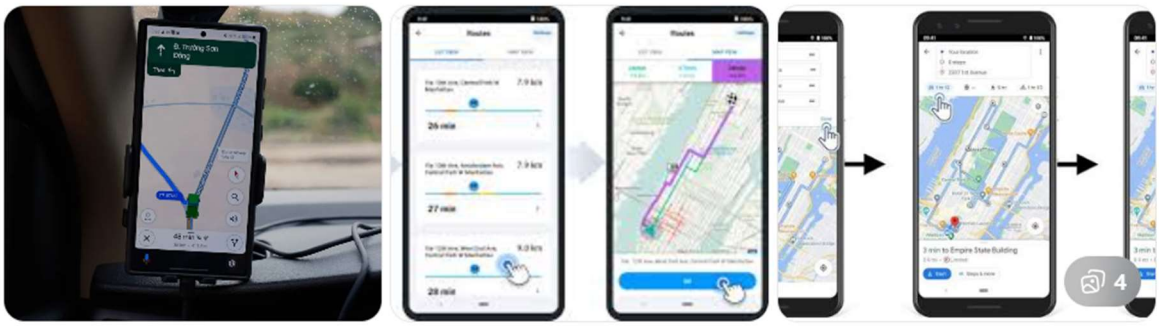

Figure 2.1: Google Maps (Laoudias et al., 2018)

One of the key strengths of GPS-based systems is their ability to provide global positioning without requiring additional infrastructure on campus. These systems work well in open spaces, provide continuous location tracking, and can integrate seamlessly with mobile devices and mapping APIs such as Google Maps (Zhang et al., 2017; Li et al., 2017). Moreover, GPS-based navigation

systems allow real-time route recalculation in response to user movement, ensuring dynamic guidance for students and staff.

However, GPS-based systems face significant limitations indoors. Satellite signals are often weak or blocked by walls, resulting in inaccurate location estimates (Laoudias et al., 2018; Zafari, Gkelias, & Leung, 2019). Urban canyons, dense building clusters, or large indoor facilities further degrade accuracy. This makes GPS-based systems unsuitable for lecture halls, libraries, or administrative offices where students require precise indoor directions.

In terms of suitability, GPS-based systems are ideal for outdoor navigation across large university campuses or multi-building complexes. They are best used for guiding students between buildings or from campus entrances to specific locations. For indoor guidance, GPS-based systems need to be complemented with alternative technologies, such as Wi-Fi fingerprinting or Bluetooth beacons, to maintain continuity and precision (Cheng et al., 2016; Li et al., 2017).

2.4.2 Indoor Positioning Systems (IPS)

Indoor Positioning Systems (IPS) are designed to locate users within buildings where GPS signals are unavailable or unreliable. IPS uses technologies such as Wi-Fi triangulation, Bluetooth Low Energy (BLE) beacons, RFID, or Ultra-Wideband (UWB) to determine positions with high accuracy. Laoudias et al. (2018) implemented an indoor navigation system for a university campus using Wi-Fi fingerprinting, which achieved location accuracy within 1–2 meters, sufficient for room-level guidance. Borrmann et al. (2015) designed a system integrating building information modeling (BIM) data with indoor positioning to allow students to navigate complex multi-floor buildings efficiently. Figure 2.2 below shows an indoor navigation system.

Figure 2.2: Indoor navigation system (Laoudias et al., 2018)

The primary advantage of IPS is its ability to provide precise indoor location data, enabling room-level navigation and detailed path guidance. This allows students to locate specific classrooms, laboratories, or offices without confusion. Huang, Gartner, and Krisp (2018) demonstrated that indoor navigation systems with BLE beacon deployment significantly reduced wayfinding errors and improved accessibility in university buildings. Additionally, IPS can support accessibility features for users with visual impairments by providing detailed turn-by-turn directions.

However, IPS has notable limitations. Infrastructure deployment is expensive and requires ongoing maintenance, such as installing BLE beacons or calibrating Wi-Fi access points (Zafari et al., 2019; Laoudias et al., 2018). Accuracy may also vary depending on environmental factors, signal interference, or network congestion. Furthermore, IPS often requires integration with mobile applications or additional middleware, which increases system complexity compared to GPS-only systems.

Indoor positioning is most suitable for guiding students within multi-story buildings, libraries, or large academic complexes. Unlike GPS systems, IPS ensures precise indoor navigation, although it may require substantial infrastructure investment. In practice, hybrid systems that combine GPS-based outdoor navigation with IPS-based indoor guidance provide the most comprehensive solution for university campuses (Cheng et al., 2016; Li et al., 2017). Such integration allows seamless transitions between outdoor and indoor environments, ensuring continuous guidance and improved user experience. Table 2.1 below shows a GPS-based and Indoor Positioning comparision table

Table 2.1: comparing GPS-based and Indoor Positioning

Authors / Year	Comparison Aspect	GPS-Based Navigation	Indoor Positioning System (IPS)
Laoudias et al., 2018; Zafari, Gkelias, & Leung, 2019; Cheng et al., 2016	Accuracy	Suitable for outdoor navigation, less accurate indoors due to weak or blocked satellite signals	Provides room-level accuracy indoors
Borrmann et al., 2015; Laoudias et al., 2018; Huang, Gartner, & Krisp, 2018	Infrastructure Requirements	Requires no additional campus infrastructure; relies on existing GPS satellites	Requires beacons, Wi-Fi calibration, RFID tags, or UWB devices
Cheng et al., 2016; Li, Chen, & Pei, 2017	Cost	Relatively low cost since no additional infrastructure is needed	Higher installation and maintenance costs due to required hardware and calibration
Cheng et al., 2016; Li, Chen, & Pei, 2017; Huang, Gartner, & Krisp, 2018	Suitability	Ideal for outdoor navigation between buildings	Essential for indoor guidance within multi-story buildings or complex interiors
Cheng et al., 2016; Li, Chen, & Pei, 2017	Best Practice	Can be combined with IPS for hybrid systems to cover both outdoor and indoor navigation needs	Works best as part of hybrid system alongside GPS to provide seamless navigation across campus

2.5 Related Studies

Several studies have implemented smart campus navigation systems, providing insights that inform the development of the proposed system. Cheng et al. (2016) developed a hybrid campus navigation system integrating GPS for outdoor navigation and Wi-Fi for indoor guidance, achieving seamless location tracking across buildings, though the system required continuous network connectivity and frequent Wi-Fi calibration. Kim, Kim, & Yoo (2014) implemented a mobile campus map application at Seoul National University, allowing students to search for facilities and navigate outdoor spaces, but it lacked indoor positioning capabilities.

Laoudias et al. (2018) designed an indoor navigation system using Wi-Fi fingerprinting, demonstrating room-level accuracy and real-time route updates, yet the system was limited by high infrastructure requirements and potential signal interference. Similarly, Borrmann et al. (2015) integrated Building Information Modeling (BIM) with indoor navigation, allowing users to locate offices and lecture halls precisely; however, scalability across large campuses remained challenging. Li, Chen, & Pei (2017) implemented a hybrid system combining GPS and indoor positioning sensors with dynamic route recalculation, which improved navigation reliability but required extensive sensor deployment.

Zafari, Gkelias, & Leung (2019) reviewed indoor positioning technologies and highlighted the effectiveness of BLE beacons and Wi-Fi triangulation in university campuses, but noted that maintenance and infrastructure costs could be prohibitive. Huang, Gartner, & Krisp (2018) implemented an indoor navigation prototype incorporating semantic search for functional spaces (e.g., library, lab), which enhanced wayfinding for students but did not fully integrate outdoor navigation. Castleman & Page (2015) demonstrated that automated reminders and notifications improve student compliance with schedules, suggesting that integrating route guidance with academic timetables can enhance system utility.

Arnold & Pistilli (2012) implemented the Course Signals system at Purdue University to track student engagement and provide early guidance, highlighting the value of linking navigation systems to academic progress. Okonkwo & Ade-Ibijola (2021) reviewed educational chatbots and navigation systems, emphasizing that real-time guidance and interactive features improve student satisfaction, but noted that few systems are fully hybrid or inclusive of accessibility features.

These studies collectively indicate that while GPS-based and indoor positioning systems have been successfully implemented in various universities, gaps remain in seamless hybrid integration, offline functionality, accessibility support, and schedule-aware guidance. The proposed Smart Campus Navigation System at ABC University seeks to address these limitations by combining outdoor GPS and indoor positioning, providing real-time dynamic routing, supporting offline access, and integrating academic timetables for more inclusive and efficient campus navigation. Table 2.2 below shows a literature review matrix of key studies.

Table 2.2. shows a summary of key studies on smart campus navigation systems

Author(s) & Year	System/Project	Functions	Features	Components	Type (GPS/Indoor/ Hybrid)	Key Findings	Limitations /Gaps
Cheng et al., 2016	Hybrid Campus Navigation	Outdoor & indoor navigation	Real-time updates, route recalculation	GPS, Wi-Fi IPS, Routing Engine	Hybrid	Continuous guidance, improved accuracy	Requires Wi-Fi calibration, high maintenance
Kim, Kim, & Yoo, 2014	Mobile Campus Map	Search and locate facilities	Mobile-friendly, real-time search	Database, GPS	GPS-based	User-friendly outdoor navigation	Does not support indoor guidance
Laoudias et al., 2018	Indoor Navigation	Indoor positioning	Wi-Fi fingerprinting, real-time updates	IPS, Routing Engine	Indoor	Room-level accuracy	High infrastructure cost, limited scalability

2.6 Identified Research Gaps

Despite the growing number of smart campus navigation systems, several limitations remain in existing implementations. Most GPS-based systems provide accurate outdoor navigation but fail to support indoor guidance due to weak satellite signals, while indoor positioning systems such as Wi-Fi or BLE beacon-based solutions require significant infrastructure, which is often costly and not widely deployed (Laoudias et al., 2018; Borrmann et al., 2015). Consequently, users frequently experience a lack of seamless navigation when moving between outdoor and indoor environments.

Another limitation observed in current systems is the absence of real-time dynamic route adjustments. Many applications rely on static maps or pre-calculated paths, which do not account for temporary building closures, events, or user deviations, reducing reliability and user trust (Cheng et al., 2016; Li, Chen, & Pei, 2017). Furthermore, most systems depend on continuous internet connectivity, limiting usability in areas with poor Wi-Fi coverage or low mobile data availability (Kim, Kim, & Yoo, 2014; Zafari, Gkelias, & Leung, 2019).

Finally, existing systems rarely consider accessibility and user-centered design features, such as support for visually impaired users, multilingual options, or personalized route preferences

(Huang, Gartner, & Krisp, 2018; Borrmann et al., 2015). In addition, few systems integrate academic schedules or class timetables, which could guide students to lecture halls and offices at the correct time. These gaps indicate the need for a hybrid, user-friendly smart campus navigation system that integrates outdoor GPS with indoor positioning, supports offline and real-time route updates, incorporates accessibility features, and aligns guidance with academic schedules to ensure efficient and inclusive campus navigation.

2.7 Summary

This chapter reviewed literature on smart campus navigation systems, highlighting their key functions, features, components, types, and gaps. Core functions include responding to queries, providing guidance, and sending real-time updates, while essential features such as real-time location tracking and search functionality. The main components are Location Detection Module and Routing Engine which enable accurate positioning and optimal path computation, supporting both GPS-based outdoor navigation and indoor positioning systems (IPS) for precise indoor guidance Despite these advancements, existing systems often lack seamless indoor-outdoor integration, real-time route adjustment, offline functionality, and accessibility or schedule-aware features.These insights provide the foundation for developing a hybrid, user-friendly Smart Campus Navigation System that addresses these limitations and ensures reliable, inclusive, and continuous guidance across ABC University's campus.